



Cybersecurity

Networking II Challenge Submission File

In a Network Far, Far Away!

Make a copy of this document to work in, and then for each mission, add the solution below the prompt. Save and submit this completed file as your Challenge deliverable.

Mission 1

1. Mail servers for starwars.com:

```
sysadmin@vm-image-ubuntu-dev-1:~$ nslookup -type=mx starwars.com
Server:          8.8.8.8
Address:         8.8.8.8#53
```

Non-authoritative answer:

```
starwars.com      mail exchanger = 1 aspmx.l.google.com.
starwars.com      mail exchanger = 5 alt1.aspx.l.google.com.
starwars.com      mail exchanger = 10 aspmx2.googlemail.com.
starwars.com      mail exchanger = 10 aspmx3.googlemail.com.
starwars.com      mail exchanger = 5 alt2.aspmx.l.google.com.
```

2. Explain why the Resistance isn't receiving any emails:

The new primary mail server is asltx.1.google.com, and the secondary should be asltx.2.google.com. In the above MX command these 2 mail servers are not listed at all and are incorrect.

3. Suggested DNS corrections:

Correct the IP address and mail servers to restore connection.

Mission 2

1. Sender Policy Framework (SPF) of theforce.net:

```
sysadmin@vm-image-ubuntu-dev-1:~$ nslookup -type=txt theforce.net
Server:           8.8.8.8
Address:          8.8.8.8#53

Non-authoritative answer:
theforce.net      text =
"google-site-verification=XTU_We07Cux-6WCS0Itl0c_WS29hzo92jPE341ckb0Q"
theforce.net      text = "v=spf1 a mx a:mail.wise-advice.com
mx:smtp.secureserver.net include:aspmx.googlemail.com ip4:45.63.15.159
ip4:45.63.4.215 ~all"
theforce.net      text =
"google-site-verification=ycgY7mtk2oUZMagcffhFL_Qaf8Lc9tMRkZZSuig0d6w"
```

2. Explain why the Force's emails are going to spam:

#Looking at the SPF, MX records and what mission 2's ip address should be for theforce.net, I can conclude that '45.23.176.21' does not match '45.63.15.159' and is missing mailserver: 'mailstore1.secureserver.net'

3. Suggested DNS corrections:

Correct the IP address and mail servers to restore connection.

Mission 3

1. Document the CNAME records:

```
sysadmin@vm-image-ubuntu-dev-1:~$ nslookup -type=cname www.theforce.net
Server:           8.8.8.8
Address:          8.8.8.8#53
```

Non-authoritative answer:

www.theforce.net canonical name = theforce.net.

2. Explain why the subpage `resistance.theforce.net` isn't redirecting to `theforce.net`:

When using cname we see that the canonical name is theforce.net and not the above mentioned domain which causes the redirect not to work.

3. Suggested DNS corrections:

Correct canonical name theforce.net to resistance.theforce.net

Mission 4

1. Confirm the DNS records for `princessleia.site`:

```
sysadmin@vm-image-ubuntu-dev-1:~$ nslookup -type=ns princessleia.site
Server:           8.8.8.8
Address:          8.8.8.8#53
```

Non-authoritative answer:

princessleia.site nameserver = ns25.domaincontrol.com.

princessleia.site nameserver = ns26.domaincontrol.com.

2. Suggested DNS record corrections to prevent the issue from occurring again:

Mission 4 states that `ns2.galaxybackup.com` is the backup, but is not listed as a nameserver. To fix the issue we would add `ns2.galaxybackup.com` to the nameserver lists.

Mission 5

1. Document the shortest OSPF path from Batuu to Jedha:

a. OSPF path:

Batuu, d,c,e,f,j,i,l,q,t,v, Jedha

b. OSPF path cost:

1,2,1,1,1,1,6,4,2,2,2 (23 total cost)

Mission 6

1. Wireless key:

Command used: aircrack-ng Darkside.pcap -w /usr/share/wordlists/rockyou.txt
Results:

KEY FOUND! [dictionary]

Master Key : 5D F9 20 B5 48 1E D7 05 38 DD 5F D0 24 23 D7 E2
52 22 05 FE EE BB 97 4C AD 08 A5 2B 56 13 ED E2

2. Host IP addresses and MAC addresses:

a. Sender MAC address:

00:13:ce:55:98:ef

b. Sender IP address:

172.16.0.101

c. Target MAC address:

00:0f:66:e3:e4:01

d. Target IP address:

172.16.0.1

Mission 7

1. Screenshot of results:

